Supplementary Document

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Study scope: This document summarizes an experiment-scoped Research Atlas with source-traceable results.

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1 Supplement Reader Crosswalk

This crosswalk explains how the supplement should be read with the main article. Each row identifies the main article surface, the supplement section that supplies the detailed audit evidence, the primary source artifact, and the stronger claim that remains beyond this study. The table is a navigation aid; it does not add experiments or promote a method.

Main article surface	Supplement support	Primary source artifact	Stronger claim requiring validation
CQR, Mondrian, and CV+ descriptive performance	S1 robustness diagnostics, S1b CQR backend sensitivity, and S2 post-selection diagnostics	`method_selection_robustness_audit; cqr_fixed_vs_model_matched_synthesis`	Method-choice, global best-method, and deployment readings would need separate validation studies.
Venn-Abers bridge negative evidence	S2 bridge validation slice and S6 traceability	`dataset_final_gate_post_selection_validation_bridge_results`	No validated Venn-Abers regression interval claim.
Bounded-support validity	S3 bounded-support endpoint policy	`bounded_support_endpoint_closure_audit`	No endpoint or bounded-support validity claim.
Population-level group inference	S4 group diagnostics	`group inference_population_readiness_audit`	No population-level group inference claim.
Duplicate and leakage integrity	S5 duplicate and integrity caveats	`cross_run_integrity_audit`	No claim strengthening by hiding duplicate or split-integrity caveats.
Knowledge-graph traceability	S6 traceability	`knowledge_graph_quality_summary`	The KG and site support navigation and source tracing.

2 Supplement Reading Protocol

The supplement is intentionally broader than the main article. Each section adds audit detail, but the extra detail must be read through a specific boundary. The protocol below states what a reviewer should check, which evidence metric anchors the check, the allowed use of that evidence, and the overclaim that remains beyond this study.

Section	Reviewer check	Evidence metric	Allowed use	Beyond this study use
S1	Check whether the observed method pattern is stable enough to describe.	CQR common-cell wins 58 of 94 cells; bootstrap selection count is cqr=1,000.	Report the fixed-GBM CQR, Mondrian, and CV+ pattern as experiment-scoped evidence.	Study-wide method choice or universal deployment use would require a separate validation study.
S1b	Check whether CQR's signal survives a backend-confound diagnostic.	Model-matched CQR completed 4,564 rows; paired cells=224; selected cells are fixed_gbm_cqr=116, model_matched_cqr=71, neither=37.	Use as a backend sensitivity check for the experiment-scoped CQR signal.	Use the check as backend sensitivity evidence rather than a universal CQR deployment rule.
S1c	Check the completed balanced Benchmark v2 result surface.	Benchmark v2 terminal rows=42,000; completed rows=29,683; selected cells are cqr_model_matched=4,415, cv_plus=1,216, mondrian_abs=1,265, split_abs=281, jackknife_plus=173.	Use as balanced follow-up evidence under the frozen Benchmark v2 utility rule.	Do not turn Benchmark v2 selections into a universal method-choice or deployment rule.
S2	Check whether post-selection and bridge diagnostics preserve the same boundary.	Post-selection diagnostic counts are cqr=18 and mondrian_abs=7; the bridge slice has 45 completed atomic runs.	Use the slice as diagnostic stress-test evidence.	A study-wide method choice or validated Venn-Abers regression interval would require a separate validation study.
S3	Check whether bounded-support endpoint behavior supports a validity claim.	0 bounded-support-validity-ready bundles and 11 raw endpoint-excursion bundles.	Report endpoint policy closure and retained domain caveats.	Bounded-support or endpoint validity would require a separate validation study.
S4	Check whether group diagnostics define a population-inference estimand.	187 pairwise group comparisons, 0 population-inference-ready bundles, and 0 population-estimand-declared bundles.	Use group rows as heterogeneity diagnostics.	Population-level group inference or deployment group inference would require a separate validation study.
S5	Check whether duplicate and leakage controls strengthen or limit interpretation.	29 duplicate actions, 46 quarantined actions, 0 unquarantined actions, and hard leakage not detected in scanned artifacts.	Retain integrity caveats as part of the scientific record.	Integrity caveats remain part of the scientific record.
S6	Check whether the evidence map helps readers trace source support.	3,643 KG nodes, 21,019 edges, 0 isolated nodes, with public source summaries available.	Use the KG and package for claim tracing and source navigation.	KG structure supports evidence tracing rather than new empirical evidence.

3 S1. Method Selection Robustness Diagnostics

This section supports the main article by separating descriptive method behavior from study-wide method choice. In conformal prediction, `1 - alpha` is the target coverage level and `alpha` is the target miscoverage rate [3]. CQR builds an interval from lower and upper quantile models before calibration [7], while CV+ and jackknife-style methods use out-of-fold or leave-one-out predictions [1, 2].

Diagnostic item	Value	Scope
Candidate methods	3	CQR, CV+, and Mondrian absolute-residual calibration only
Common dataset-alpha cells	94	Shared comparison cells
Common-cell selected diagnostic method	`cqr`	Retrospective diagnostic only
Common-cell counts	cqr=58; cv_plus=15; mondrian_abs=21	Descriptive counts only
Bootstrap replicates	1,000	Selection stability diagnostic
Bootstrap counts	cqr=1,000	Experiment-scoped diagnostic
Bootstrap primary selection rate	1.0000	Stability estimate for the retrospective pattern
Bootstrap primary selection-rate descriptive range	0.9962 to 1.0000	Precision check for the diagnostic rate
Leave-one-dataset retention	1.0000	Diagnostic robustness
Leave-one-alpha retention	1.0000	Diagnostic robustness
Study-wide method choice	`beyond this study`	Requires prospective validation rather than retrospective diagnostics
Pairwise shared cells	94	Inferential support within the stated scope
Main-result primary win rate	0.6778	Descriptive rate for the audited comparison
Main-result primary win-rate descriptive range	0.5757 to 0.7653	Uncertainty around the descriptive rate
Robustness common-cell primary win rate	0.6170	Common-cell diagnostic only
Robustness common-cell descriptive range	0.5160 to 0.7089	Precision check for the audited scope

Reader note: this section reports the fixed-GBM CQR pipeline, Mondrian calibration, and CV+ as experiment-scoped descriptive signals. The descriptive ranges summarize uncertainty around the observed rates; they do not turn the retrospective comparison into a method-choice result.

3.1 S1b. CQR Backend Sensitivity Check

The completed model-matched CQR rerun checks whether the historical fixed-GBM CQR backend explains the CQR signal. It preserves the experiment-scoped interpretation and does not turn the retrospective comparison into a study-wide method choice.

Backend diagnostic item	Value	Scope
Fixed-GBM CQR completed rows	4,564	Historical comparator rows
Model-matched CQR completed rows	4,564	Completed rerun rows
Paired dataset-alpha-model-family cells	224	Direct CQR backend comparison cells
Fixed-GBM CQR selected cells	116	Coverage-eligible interval-score selections
Model-matched CQR selected cells	71	Coverage-eligible interval-score selections
Neither coverage-eligible variant	37	Cells where both CQR variants fail the coverage-eligibility rule
Study-wide method choice supported	`False`	Backend sensitivity evidence only

3.2 S1c. Benchmark v2 Completion Check

Benchmark v2 is the balanced follow-up benchmark used to separate the retrospective result surface from a frozen, paired execution surface. It uses the frozen coverage-tolerance gate followed by interval score. The table below is descriptive follow-up evidence, not a method-choice or deployment rule.

Benchmark v2 item	Value	Scope
Planned method rows	42,000	Frozen execution surface
Terminal method rows	42,000	Complete accounting; no pending rows
Completed method rows	29,683	Rows with fitted interval results
Failed method rows	0	Execution failure count
Paired cells	8,400	Dataset-task-seed-learner comparison cells
Selected paired cells	7,350	Coverage-eligible selected cells under the frozen rule
CQR model-matched selections	4,415	Largest Benchmark v2 selected-cell signal
CV+ selections	1,216	Secondary selected-cell signal
Mondrian selections	1,265	Secondary selected-cell signal
Split conformal selections	281	Baseline selected-cell signal
Jackknife+ selections	173	Plus-family selected-cell signal under caps/skips
Live integrity status	`pass`	Critical violations=0

Reader note: Benchmark v2 supports a clearer descriptive comparison than the earlier retrospective surface. It remains a benchmark-level comparison rather than a rule for every future application. The supported wording is that model-matched CQR was selected most often under the frozen Benchmark v2 rule, with CV+ and Mondrian as important comparators.

4 S2. Post-Selection Validation Diagnostics

Post-selection validation asks whether the descriptive pattern persists in a held-out validation-style slice after the candidate methods have already been identified. The supplement treats this as a diagnostic stress test. It does not convert the observed counts into a study-wide method choice.

Diagnostic item	Value	Scope
Validation datasets	5	Narrow validation scope
Validation dataset-alpha cells	25	Shared cells only
Completed validation atomic runs	225	Completed diagnostic rows
Validation diagnostic counts	cqr=18; mondrian_abs=7	No study-wide method choice
Validation primary win rate	0.7200	Post-selection diagnostic rate
Validation primary win-rate descriptive range	0.5242 to 0.8572	Wide uncertainty remains visible
Feature leakage violations	0	Leakage sidecars clean in this slice
Width pathology rows	6	Retained caveat evidence
Bridge validation datasets	1	Dataset-final-gate bridge slice
Bridge validation atomic runs	45	Auxiliary diagnostic evidence
Bridge validation diagnostic counts	cqr=3; cv_plus=1; mondrian_abs=1	Bridge-slice diagnostics only
Venn-Abers bridge under-coverage runs	14	Negative bridge evidence

Reader note: post-selection diagnostics are stress tests of a descriptive pattern. They do not justify language such as study-wide method choice, globally best conformal method, or validated Venn-Abers regression interval. The Venn-Abers rows are especially narrow: they concern the evaluated bridge, while predictive-distribution and generalized Venn-Abers work remain separate literature objects [4, 5, 9, 6].

5 S3. Bounded-Support Endpoint Policy

Bounded-support checks ask whether prediction interval endpoints respect the natural domain of the target. This is different from ordinary marginal coverage. Weighted conformal ideas can address some covariate-shift settings [8], but the present endpoint policy is an empirical domain-validity diagnostic. The current evidence does not support bounded-support validity claims.

Diagnostic item	Value	Scope
Bounded-support bundles	15	Research Document-candidate bundles
Bounded-support datasets	6	Current endpoint-audit scope
Raw endpoint-excursion bundles	11	Domain limitation retained
Post-handling validated bundles	15	Policy triage complete
Clean or not-applicable bundles	4	Not enough for global validity
Validity-ready bundles	0	No bounded-support validity claim
stronger claim-ready bundles	0	No positive endpoint claim
Endpoint support status counts	beyond this study__natural__domain__endpoint__excursions=11; clean__no__natural__domain__endpoint__excursions=3; not__applicable__unbounded__target__endpoint__hygiene__recorded=1	Support-status inventory, not a validity upgrade

Reader note: bounded-support evidence is stricter than ordinary coverage evidence. A method can have acceptable empirical coverage and still fail a natural-domain endpoint gate. The supplement therefore reports endpoint closure as a limitation, not as a post-processing success story.

6 S4. Group Diagnostics

Group coverage diagnostics compare interval behavior across observed groups. They are useful for auditing heterogeneity, but they are not population-level group inference claims unless the population estimand, protected-attribute scope, sampling design, and multiplicity treatment support that claim.

Diagnostic item	Value	Scope
Group diagnostic bundles	15	Diagnostic group scope
Group diagnostic datasets	6	Current diagnostic scope
Bootstrap replicates	500	Gap-uncertainty diagnostic
Group counts recorded	15	Required diagnostic metadata
Coverage by group recorded	15	Group comparison evidence
Width by group recorded	15	Interval-size comparison evidence
Gap uncertainty recorded	15	Multiplicity-aware caveat evidence
Pairwise group comparisons	187	Multiplicity scope, not group-inference proof
Population-estimand-declared bundles	0	Required for population-level group inference, absent here
Protected-attribute-scope-declared bundles	0	Required scope declaration, absent here
Weighted-estimand-applied bundles	0	No weighted population estimand applied
Sampling-weight-policy-declared bundles	15	Policy exists but does not open group-inference claim
Population-group-inference-ready bundles	0	No population-level group inference claim

Reader note: group rows can show where interval behavior differs, but group inference is an estimand-level claim. Because the current population estimand and protected-attribute scope are not declared at the bundle level, the correct supplement reading is diagnostic heterogeneity, not population-level group inference.

7 S5. Duplicate And Integrity Caveats

Duplicate and split-integrity checks protect the interpretation of model comparisons. If repeated rows, row signatures, or model-visible duplicates can affect train/calibration/test separation, the result must retain the caveat rather than become stronger prose.

Diagnostic item	Value	Scope
Duplicate actions	29	Caveat inventory scope
Open duplicate actions	0	Closure evidence only
Paired duplicate comparison rows	1,043	Sensitivity evidence
Quarantined actions	46	Stronger-claim validation retained
Unquarantined actions	0	No untracked duplicate action
Cross-run leakage status	`hard_leakage_not_detected_in_scanned_artifacts`	Hard leakage not detected in scanned artifacts
Unsupported claim hits	0	Claim scan clean for this audit
Cross-run risk counts	medium=39; pass=109	Risk inventory retained
Cross-run caveat counts	duplicate_cluster_plus_family_internal_fold_caveat=17; duplicate_signature_cross_split_caveat=14; model_visible_signature_cross_split_caveat=15	Plus-family rows are read with the fold-local preprocessing caveat; other caveats constrain duplicate and cross-split interpretation.

Reader note: the clean unsupported-claim scan does not erase duplicate or split caveats. It means the scanned artifacts did not contain unsupported claim language; the caveats still travel with the empirical interpretation.

8 S6. Traceability

The supplement is linked to a knowledge graph snapshot with 3,643 nodes, 21,019 edges, and 0 isolated nodes. The graph connects report sections, source artifacts, citations, study limits, and quality checks. Use it as an evidence map for navigating the Research Atlas.

Venn-Abers evidence is retained as part of the negative and boundary evidence for this study [4]. The supplement does not claim that the current interval bridge validates Venn-Abers regression intervals.

9 Reading Guide

- This supplement records method details, robustness diagnostics, and traceability for the Research Document.
- Method evidence is experiment-scoped; use in a new setting would need its own validation.
- Benchmark v2 is a completed balanced follow-up benchmark; its selections remain descriptive.
- Bounded-support endpoint results block validity claims.

- Group diagnostics do not establish population-level group inference.
- Duplicate and integrity caveats are retained rather than hidden.

10 References

References

[1] Barber et al. (2020). *Predictive inference with the jackknife+.* `https://arxiv.org/abs/1905.02928`.

[2] Kim et al. (2020). *Predictive Inference Is Free with the Jackknife+-after-Bootstrap.* `https://arxiv.org/abs/2002.09025`.

[3] Lei et al. (2017). *Distribution-Free Predictive Inference for Regression.* `https://arxiv.org/abs/1604.04173`.

[4] Nouretdinov et al. (2018). *Inductive Venn-Abers predictive distribution.* `https://proceedings.mlr.press/v91/nouretdinov18a.html`.

[5] Nouretdinov and Gammerman (2024). *Inductive Venn-Abers Predictive Distributions: New Applications and Evaluation.* `https://proceedings.mlr.press/v230/nouretdinov24a.html`.

[6] Petej and Vovk (2026). *Inductive Venn-Abers and related regressors.* `https://arxiv.org/html/2605.06646v1`.

[7] Romano et al. (2019). *Conformalized Quantile Regression.* `https://arxiv.org/abs/1905.03222`.

[8] Tibshirani et al. (2020). *Conformal prediction under covariate shift.* `https://arxiv.org/abs/1904.06019`.

[9] Van Der Laan and Alaa (2025). *Generalized Venn and Venn-Abers Calibration with Applications in Conformal Prediction.* `https://proceedings.mlr.press/v267/van-der-laan25a.html`.

11 Source Artifacts

Detailed source paths, hashes, and public summaries are indexed in `evidence/public_artifact_manifest.json` and the Research Atlas evidence pages.